Advance Python Assignment

Topic-3: Opening and Closing Files

Q1. Explain different file modes in Python.

Ans.
In Python, files can be opened using different modes depending on the work we want to perform. The open() function is used to open a file. The 'r' mode is used to read data from a file, 'w' mode is used to write data and it creates a new file if it does not exist, and 'a' mode is used to add data at the end of an existing file. There are also modes like 'r+' and 'w+' which allow both reading and writing. Choosing the correct mode is important to avoid losing existing data or causing errors in the program.

Example:
```
f = open("demo.txt", "a")
f.write("Adding new line")
f.close()
```

Q2. Explain the use of open() and close() functions in file handling.

Ans.
The open() function in Python is used to open a file so that we can perform operations like reading or writing. Once the work is completed, the close() function is used to close the file. Closing a file is very important because it saves the changes and releases system resources. If a file is not closed properly, it may cause data loss or memory problems. Therefore, it is a good practice to always close the file after using it.

Example:
```
file = open("info.txt", "w")
file.write("File handling example")
file.close()
```